

During a visual field test, patients are instructed to look straight ahead at a fixed point as objects are flashed in their peripheral field of vision. A patient with normal vision should have:

- ✔ ☒ A. no difficulty detecting a moving point of light in the periphery in dim light conditions. [75%]
- ☐ B. no difficulty correctly identifying color presented in the periphery in dim light conditions. [8%]
- ☐ C. difficulty detecting motion in the periphery in bright light conditions. [8%]
- ☐ D. difficulty correctly identifying color presented near the point of fixation in bright light conditions. [8%]

Correct

75% answered correctly

Explanation:

The visual field	
Central (foveal)	Peripheral
• ↑ Density cone photoreceptors • Bright light conditions • Color & detail perception • ↑ Visual acuity	• ↑ Density rod photoreceptors • Dim light conditions • Motion perception • ↑ Light sensitivity

The retina is the region at the back of the eye that converts light information into neural impulses that are processed in the brain. The visual field is composed of central and peripheral vision; the [distribution of photoreceptors in the retina](#) results in differing degrees of sensitivity and acuity across the visual field.

- **Central (foveal) vision:** **Cones** are sensitive to certain wavelengths (**colors**) and high-intensity (bright) light. Cones are more densely packed at the **fovea**, the central portion of the retina that receives information from the central visual field, producing the highest **visual acuity** (image sharpness and clarity).
- **Peripheral vision:** **Rods**, present primarily in the **periphery** of the retina, are much more sensitive to light than cones. Rods do not transmit information about color but are well suited to detect **motion** and low-intensity (dim) light, producing high **visual sensitivity**.

Therefore, during a visual field test, a patient with normal vision should have no difficulty detecting a moving point of light in the periphery in dim light conditions.

(Choice B) Rods are not sensitive to color; therefore, a patient with normal vision *will* have difficulty correctly identifying color presented in the periphery in *both* bright and dim light conditions.

(Choice C) The ability to detect motion (eg, an advancing predator) in one's periphery is an

important evolutionary adaptation, and peripheral vision is designed to detect such changes, regardless of light conditions. As a result, a patient with normal vision *will not* have difficulty detecting motion in the periphery in *either* bright or dim light conditions.

(Choice D) More cones are present near/at the fovea than in the periphery. Therefore, a patient with normal vision *will not* have difficulty correctly identifying color in bright light conditions when it is presented near the point of fixation, which is within the central visual field.

Educational objective:

Central (foveal) vision is highly acute (sharp, clear) because it is produced by cone photoreceptors, which are responsible for detecting color and detail under bright light conditions. Peripheral vision is highly sensitive because it is produced by rod photoreceptors, which are responsible for detecting light and motion under dim conditions.

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Sensation, Perception, and Consciousness